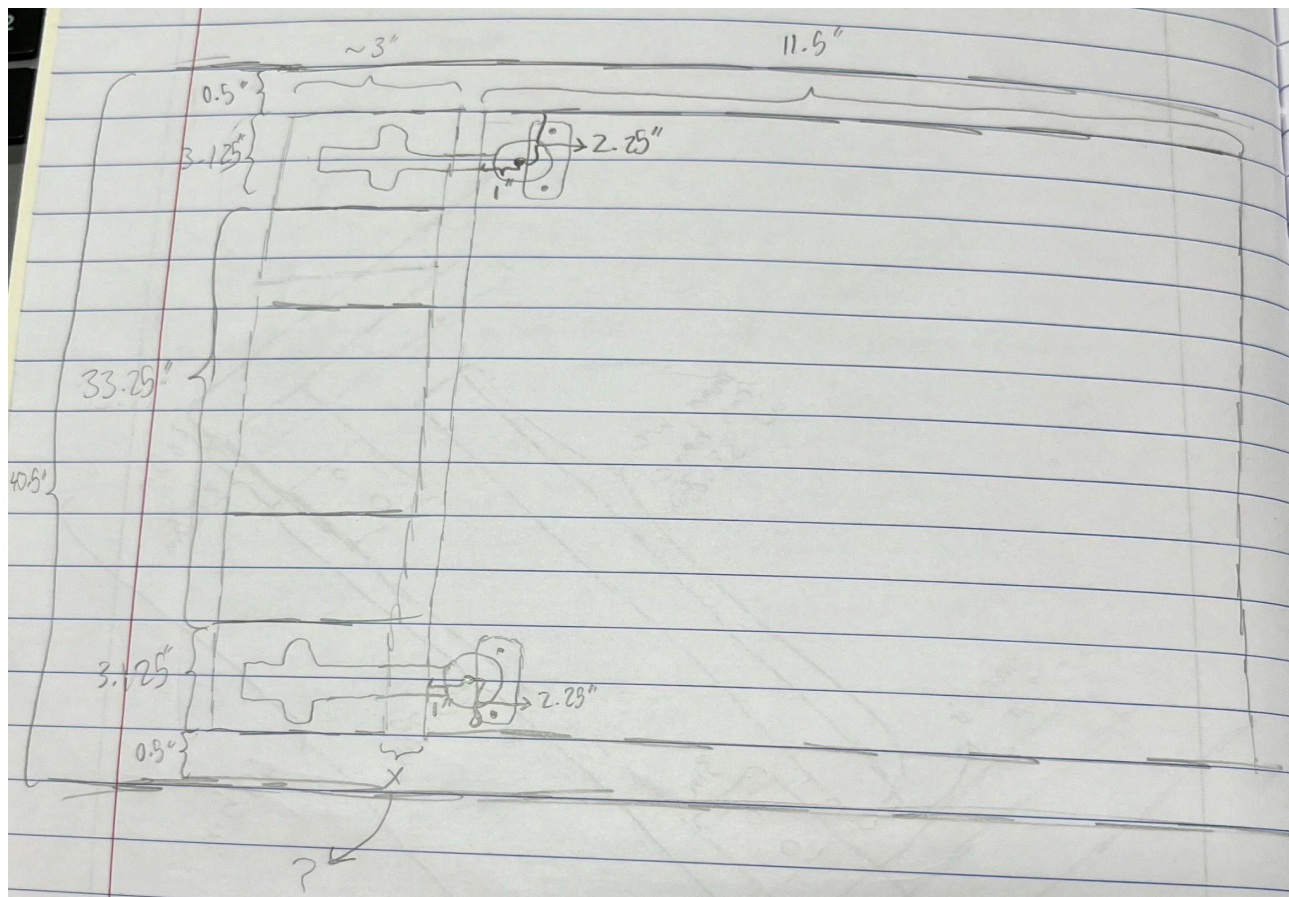


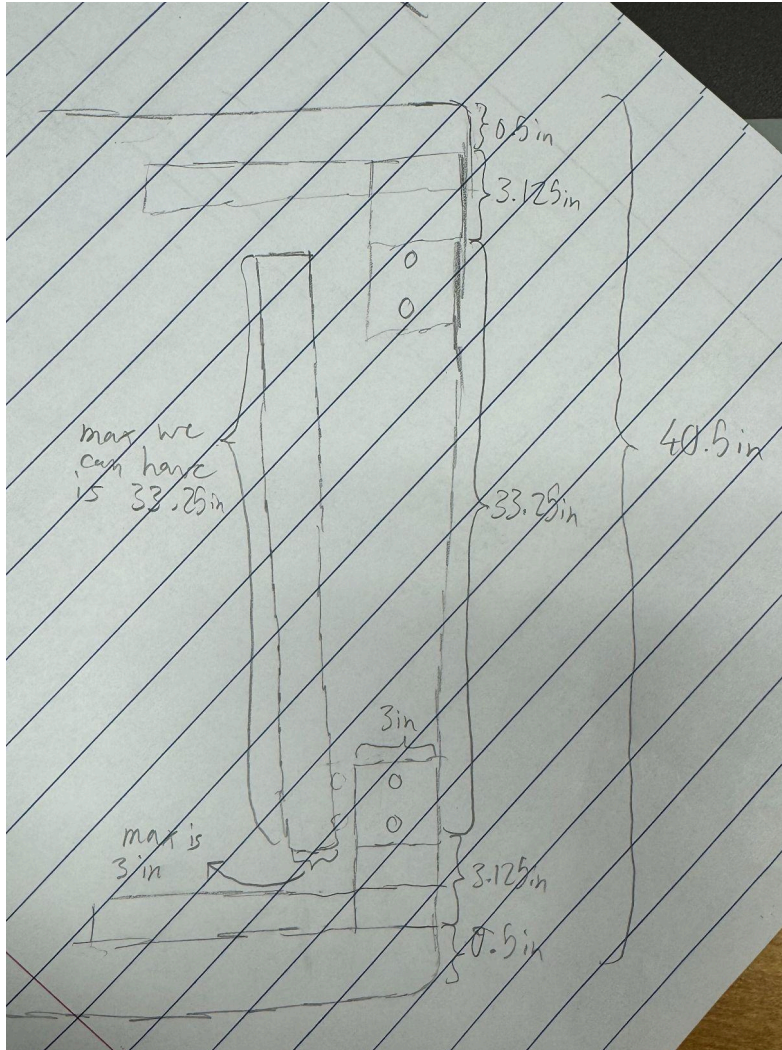
My Design: Adaptive Technology
Armin Zarbaft, Jared Wagner, John Estock

Element G: Construction of a Testable Prototype

In order to develop a prototype of a door capable of being installed in the middle island of the sensory room, we drew some blueprints with metrics to develop an understanding of the mechanics and intricacies behind each step.



This blueprint shows the metrics involved in the drilling that would allow us to attach the foundational hinge from the pocket bracket to the actual door. As seen, we needed to drill a hole for this hinge at the top and bottom of the doors that were attached to the pocket rail.



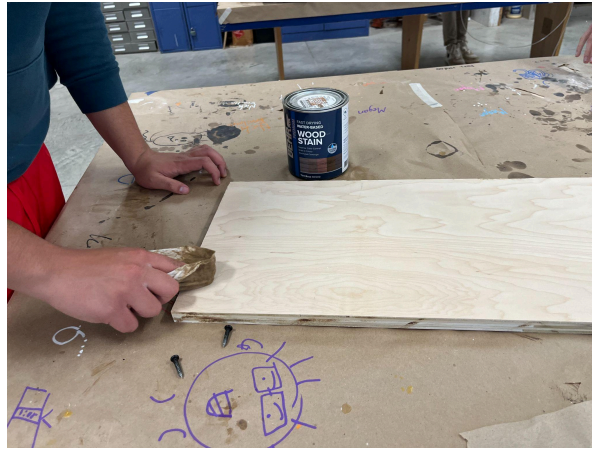
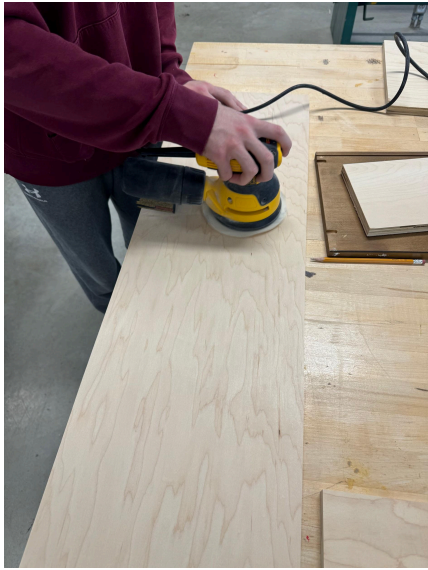
This blueprint shows the metrics needed to attach the bi-fold hinges to their respective door components. The bifold hinges allow for the door to fold onto itself, allowing for easy storage. The measurements for how far down from the top and bottom the hinges are placed is shown.

After drawing the blueprints and getting a better idea of what physically needs to be done to construct our prototype, we created a list of tasks needed to be completed. The list is not in sequential order, as we jumped around as needed to meet changing schedules and needs.

List:

- ☐ Banding
- ☐ Clear gloss stain
- ☐ Sanding
- ☐ Drilling holes for bifold hinges
- ☐ Drilling circle for rail hinges
- ☐ Developing/Buying a door handle and installing it to the door
- ☐ Buying shelf supports (and installing once at sensory room)
- ☐ Cutouts on rails
- ☐ Cutting shelves (to make room for door being stored in middle island)
- ☐ Cut attachment rails
- ☐ Additional wooden 'plank' added to door for hinges
- ☐ Cutting doors
- ☐ Colored stain
- ☐ Full installation into middle island

Images of our group completing the tasks can be seen below with their descriptions:



Colored and clear gloss staining:
In order to stain the doors to the color from the shelves, we had to first sand. After sanding all sides of the doors, we added our color-matched stain and added an additional clear gloss stain once we were done.



Cutting rails: Because we couldn't find rails short enough for the middle island, we had to cut a few inches off. We ran into the problem here of the mechanics of the sliding rail sliding out since the previous part that was stopping it had been cut off. We had to then bend part of the rail down to 'catch' the sliding part from falling out.



Drilling: After staining and adding banding, we had to drill holes in the wood for the bifold hinges for the fold-function, and circular hinges for the pocket-function. For the circular part we had to find the drillbit that fit the size of our hinge.



Shelves: Because our design had the door being pocketed in the middle island, we had to cut the previously existing shelves so that there was more room. After cutting down the shelves a few inches, we had to add supports under the shelves that could be drilled in during installation.



Prototype Final Iteration: Before installing the final product to the middle island, this is what we pieced together to demonstrate our plan's viability. Putting the pieces together upon installation has no expected issues. This prototype is expected to accomplish the functions listed:

- Hinge is attached to the door and rail
- Rail is attached to another piece of wood (referred to as additional wooden 'plank' in the list at the beginning of 'Element G')
- Door can fold onto itself and be pushed in completely.

Conclusion: We have prepared everything for installation and anticipate no issues. We managed our overall job by assigning small tasks and keeping a timeline. By assigning these small tasks, we were able to keep track of every component of construction and ensure that we didn't overlook anything. We have also kept our stakeholders informed, and are in discussions with them to start construction.